README — WATTWISE by Team DJGSS

The following files can be found in the zipped folder:

Table of Contents:

Presentation Slides [WattWise_Project_Presentation.pdf]

Slides used for presentation of the findings, proposed solutions, and a brief look of the dashboard results. The slides include detailed appendices.

Data Dictionary [WattWise_Data_Dictionary.pdf]

Detailed information regarding the proposed data to be collected and definitions of the six metrics to be tracked by the client.

Link to Dictionary:
https://1sfu-my.sharepoint.com/:b:/g/personal/jyh3_sfu_ca/IQCne9fyBWzfT79nLe25qznuAQsakYsiXvb-ShyzbrwPFc0?e=cacHcL

Sensor-Based Energy Simulation [WattWise_Code.R]

For this project, we simulated hourly energy consumption for Surrey Memorial Hospital(SMH) to evaluate the impact of sensor-based optimization (lighting and HVAC support). The model incorporates real-world operational factors (e.g., sensor failures, overrides, and missing data) and outputs datasets used to build a Power BI dashboard for different stakeholder groups.

Key Assumptions

Building & Operations

- Floor area: 115,112 m^2
- Operational intensity: 110% (high-utilization hospital)
- Electricity share: 65% of total energy use

Climate & Load Modeling

- Hourly temperature profile based on Vancouver's monthly day/night averages
- Heating and cooling are driven by:
    - Heating degree: max(18 − temp, 0)
    - Cooling degree: max(temp − 20, 0)

Sensor Deployment

- Coverage:
    - Motion sensors: ~60 m^2 per sensor
    - Daylight sensors: ~85 m^2 per sensor
    - CO_2 sensors: ~175 m^2 per sensor
    - Occupancy sensors: ~114 m^2 per sensor
- Deployment factor: 70% (targeted areas)

Energy Savings

- Lighting savings: ~45%
- HVAC savings: ~30–40%
- Combined savings applied proportionally to baseline loads

Operational Factors

- Override rate: ~3–8%
- Sensor failure rate: ~1–5%
- Missing data rate: ~1–3%

What the Code Does

1. Generate Time Series

- Creates hourly timestamps for 2025
- Extracts:
  - Hour of day
  - Day of year
  - Month

2. Simulate Temperature

- Uses monthly day/night values
- Applies a sinusoidal function to generate hourly temperature variation

3. Compute Thermal Loads

- Calculates:
  - Heating degree hours
  - Cooling degree hours
- Uses these to estimate:
  - Heating load
  - Cooling load
  - Fan energy

4. Estimate Baseline Energy

- Combines:
  - HVAC (heating, cooling, fans)
  - Lighting loads
- Produces:
  - `Facility_base` (baseline energy use)

5. Apply Sensor Savings

- Reduces lighting and HVAC loads based on assumed savings
- Produces:
  - `Facility_optimized` (ideal sensor-driven scenario)

6. Simulate Real-World Conditions

Adds operational uncertainty:

- `override_flag` → reverts to baseline
- `sensor_fail` → introduces faulty readings
- `missing_flag` → removes data points

Final output:

- `Facility_operational` (realistic energy use)

Output

The code generates a dataset with:

Column	Description
`datetime`	Hourly timestamp
`temp`	Simulated temperature
`Facility_base`	Baseline energy
`Facility_optimized`	Ideal savings scenario
`Facility_operational`	Real-world scenario
`override_flag`	Manual override indicator
`sensor_fail`	Sensor failure indicator
`missing_flag`	Missing data indicator

Output is exported as a CSV file for downstream analysis.

Key Computations

- Energy Intensity (EUI) = Total Energy (kWh) / Floor Area (m^2)
- Energy Savings = Baseline - Optimised
- CO_2e Emission Reduced (t) = (kWh Saved * 0.015) / 1000 (Based on 15 g CO_2e/kWh for BC electricity)
- Annual Savings ($): kWh Saved * 0.14 (Based on 14.06 cents per kWh - BC Hydro)
- CLIMB Points Avoided: kWh Saved / 8000 (MWh Saved / 8)

Power BI Integration [WattWise_Dashboard_Prototype.pbix]

Data Import

- Load the CSV into Power BI
- Ensure:
    - datetime is parsed correctly
    - Numeric fields are properly typed

Measures Created (DAX)

- Total Energy Saved
- Annual Cost Savings
- CO_2 Reduction
- CLIMB Points Avoided
- EUI Reduction
- Override Rate
- Sensor Failure Rate
- Missing Data Rate

Dashboard Structure

Executives

- Energy saved
- Cost savings
- CO_2 reduction
- CLIMB points

Site Managers

- EUI reduction
- Override rate
- Sensor reliability

Analysts

- Hourly load breakdown (heating, cooling, fans, lighting)
- Missing data trends

Dashboard [WattWise_Dashboard_Overview.pdf]

A static PDF version of the interactive Power BI dashboard.